



Ecological – Agricultural – Cultural Integration of Border Villages from a Post-Productivist Perspective

A Case Study of Lai Chi Wo Village, Hong Kong

AUTHOR

Shuxuan Duan
Shengyu Zhao
Ying Li
Yao Zhang
Yuxin Zhang
Tianlun Yi
Jing Zhong

Tongji University
Fudan University
Tsinghua University
Tsinghua University
Tsinghua University
Tsinghua University
Tsinghua University

2430317@tongji.edu.cn
24210500006@m.fudan.edu.cn
liying24@mails.tsinghua.edu.cn
Yao-Zhan24@mails.tsinghua.edu.cn
2634477548@qq.com
yiti22@mails.tsinghua.edu.cn
zhongji24@mails.tsinghua.edu.cn

Ecological–Agricultural–Cultural Integration of Border Villages from a Post-Productivist Perspective: A Case Study of Lai Chi Wo Village, Hong Kong

Shuxuan Duan¹; Shengyu Zhao²; Ying Li³; Yao Zhang⁴; Yuxin Zhang⁵; Tianlun Yi⁶; Jing Zhong^{*}

¹Master of Architecture, College of Architecture and Urban Planning, Tongji University

²Master of business management, Department of Tourism, Fudan University

³Master of Public Management, School of Public Policy & Management, Tsinghua University

⁴Master of Urban and Rural Planning, School of Architecture, Tsinghua University⁵Yuxin Zhang,

⁵Master of Architecture, Shenzhen International Graduate School, Tsinghua University

⁶PhD in Data Science and Information Technology, Shenzhen International Graduate School, Tsinghua University

^{*}Master of Architecture, Shenzhen International Graduate School, Tsinghua University

Abstract

In the Guangdong–Hong Kong–Macao Greater Bay Area, tourism in the Hong Kong–Shenzhen border region is becoming more diverse and integrated. This study examines rural revitalization and tourism in this region, focusing on Lai Chi Wo Village. The village faced population out-migration in the 1950s but has seen restoration efforts since 2000. However, it still faces challenges like population constraints and limited financial resources. Using a post-productivist perspective, the study integrates the agriculture-cultural-tourism model with "the sixth industrialization of agriculture" strategy to build practical pathways. It emphasizes collaborative governance and collective action to transform the border area into a vibrant hub. The study proposes recommendations for infrastructure upgrading, community cultivation, product development, and financial operations, offering guidance for the sustainable development of border villages in the Hong Kong–Shenzhen region.

Keywords: border tourism; Hong Kong; rural revitalization; agriculture-cultural-tourism integration; collaborative governance

1.Introduction

Lai Chi Wo is located in the northeastern part of Xinjie Area, near Plover Cove Country Park and Yan Chau Tong Marine Park. As a remote suburban area of Hong Kong and a border zone between Shenzhen and Hong Kong (Figure 1.1), the village has historically been designated a restricted area by the Hong Kong Special Administrative Region government to prevent illegal immigration. Entry is permitted only with prior approval.



Fig.1.1 Lai Chi Wo location map

Lai Chi Wo, a historic Hakka village with a history spanning over 400 years, witnessed significant depopulation in the 20th century as villagers migrated for work. Since 2000, some villagers returned to restore old houses and catalyzed the involvement of multiple stakeholders, including government agencies, social organizations, universities and enterprises. This collaborative effort has revitalized ecological farmland, preserved the traditional village, and fostered integrated ecological, cultural, social, and economic development (Figure 1.2). The project was recognized with the United Nations Sustainable Development Contribution Award in 2020 and the Asia Urban Landscape Award in 2021. Despite these accolades, sustaining the project and achieving self-maintenance remain critical challenges for its future.

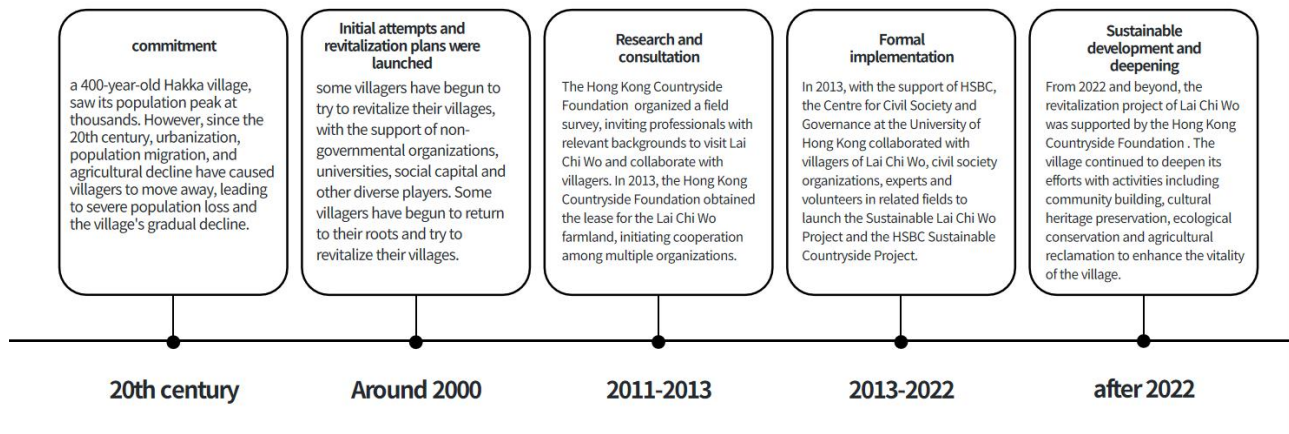


Fig.1.2 Timeline of rural revitalization in Lai Chi Wo

2.Methods: Establishing a "Post-Productivism – Agri-Cultural-Tourism – Collaborative Governance" Analytical Framework

The theoretical framework proposed in this study consists of three levels. At the macro level, the concept of post-productivism is introduced to explain the collective dilemmas faced by rural areas globally. At the meso level, the agriculture-cultural-tourism (ACT) integration model, along with "the sixth industrialization of agriculture" strategy, serves as a guiding framework for rural revitalization practices. At the micro level, the focus is on collaborative governance and collective action mechanisms, translating macro concepts and meso-level pathways into practical implementation. This framework not only enables an in-depth analysis of the underlying causes of rural decline in the Sha Tau Kok area of Hong Kong, but also provides a systematic set of guidelines for its future sustainable development (Figure 2.1).

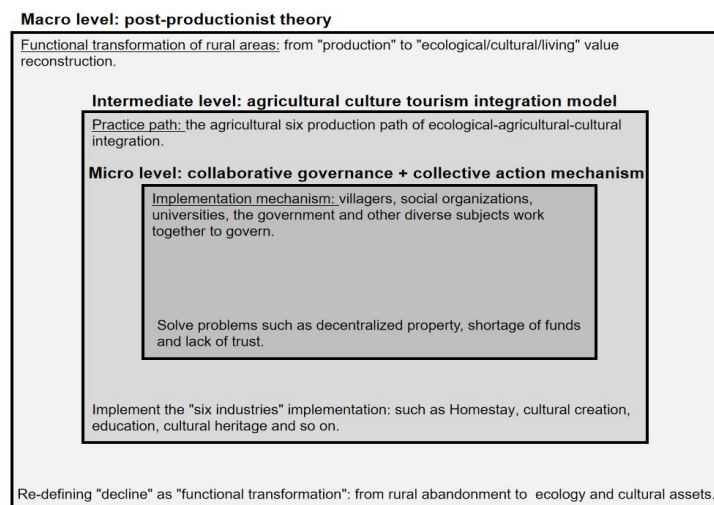


Fig.2.1 Theoretical framework diagram

2.1 Macro Perspective: Post-Productivism Theory

Rural decline often accompanies industrialization and urbanization. In the "productivist" era, rural areas mainly provided cities with agricultural products, labor, and raw materials, emphasizing food production. However, as societies reach a prosperous "mature postmodern" stage, values and economic structures change. Post-productivism theory emerged in this context. It challenges the view that production is the only key function of rural areas, suggesting rural spaces should also meet society's needs for ecological conservation, cultural landscapes, and recreational services. In Hong Kong, a highly developed economy, rural population outflow and land abandonment result from the shift from productivism to post-productivism.^[6] When villagers no longer depend on farming, other elements like physical spaces, Hakka cultural heritage, and natural ecology become important for future rural revitalization.

2.2 Meso Perspective: Agriculture-Cultural-Tourism Integration Theory and "the sixth industrialization of agriculture" Strategy

The ACT integration model offers a way to achieve multifunctional rural objectives ^[5]. It combines agriculture, culture, and tourism into a synergistic industrial ecosystem. Agriculture provides landscapes and products, culture adds historical and humanistic elements to villages, and tourism turns these resources into experiences for visitors. The essence of ACT integration is mutual industry penetration, enabling the transformation of traditional farming into cultural experiences and achieving reverse integration. This has created new rural business forms, breaking industry boundaries and optimizing resource allocation. The "sixth industrialization of agriculture" strategy deepens the ACT model and guides industrial development in Sha Tau Kok. It extends the agricultural value chain beyond production to include secondary, tertiary, and higher-level functions, enhancing rural value.^[2] (Figure 2.2).

2.3 Micro Mechanisms: Dynamic Integration of Collaborative Governance and Collective Action

Rural social network theories and paths need effective micro-level mechanisms for realization. Collaborative governance and collective action theory bridge theory and practice. In Hong Kong with highly privatized land, rural revitalization needs multi-stakeholder governance rather than relying on a single actor. Collaborative governance surpasses traditional government-led models, emphasizing coordination, cooperation, and joint decision-making among multiple actors for public goals. The success of Lai Chi Wo exemplifies this model. During its revitalization, social organizations like the Hong Kong Countryside Foundation and the University of Hong Kong provided leadership, expertise, and resources; enterprises such as HSBC and the Hong Kong Jockey Club offered initial funding; the government later provided policy support and infrastructure; and villagers participated in decision-making via community organizations like "Pei Shing Tong" and "HakkaHome" to ensure community agency^[4](Figure 2.3).

Within the "action arena" of collaborative governance, collective action theory offers a micro-level logic for analyzing interactions among stakeholders. It illustrates how to foster cooperation and surmount challenges such as free-riding in situations of conflicting private interests and low trust, by establishing mechanisms of trust, reputation, and reciprocity. In Lai Chi Wo, problems such as fragmented land ownership and ideological differences in farmland restoration and rural accommodation projects exemplify typical collective action dilemmas. Thanks to the intervention of leaders like Lam Chiu Ying, ongoing communication with villagers, and the tangible benefits of the projects, stakeholders gradually developed mutual trust, facilitating coordinated collective action^[10].

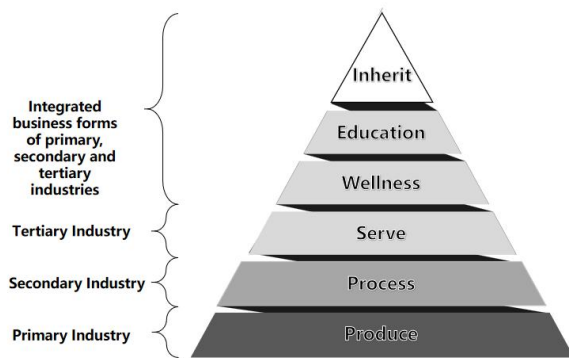


Fig.2.2 Diagram of the interlinkages in "the sixth industrialization of agriculture"

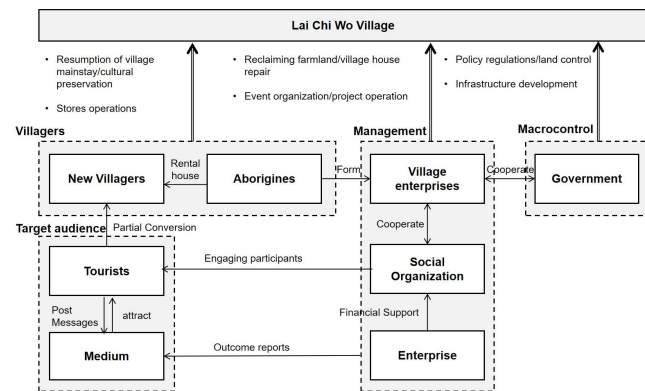


Fig2.3 Diagram of the multi-stakeholder collaborative governance system in Lai Chi Wo

3. Development challenges

3.1 Limited Accessibility Due to Geographical Isolation

Transportation serves as the lifeline of rural areas and supports socio-economic activities. The geographical isolation of Lai Chi Wo acts as a bottleneck to its development. Traditional land-based transport cannot reach the village directly. The government constructed natural trails primarily for hikers. The primary mode of public transport is the public ferry service. However, ferry services are limited and can be easily disrupted by inclement weather, posing challenges to daily commuting, goods transportation, and emergency situations. This isolation increases living costs and travel time, making it difficult to attract daily commuters and casual tourists. For potential young returnees, it signifies a disconnection from urban opportunities and a high threshold for resettlement. Limited accessibility leads to challenges in population influx, a narrow visitor market, and restricted industrial development^[3]. Without enhancements to the transportation infrastructure, efforts to "revitalize" Lai Chi Wo are constrained, and deep integration with the external system is difficult.

3.2 Cultural Heritage Challenges and Physical Space Decay

The core appeal of Hakka villages lies in their rich cultural heritage and well-preserved settlement morphology, yet both are confronted with significant challenges. Culturally, as the older generation ages or moves away, there is a growing gap in the transmission of intangible cultural heritage, such as orally-transmitted history, traditional skills, and ritual festivals. The younger generation, frequently from urban or overseas backgrounds, often lacks a sense of cultural identity and motivation to preserve these traditions. As of January 2025, Lai Chi Wo had only 30 permanent residents, predominantly elderly and a few "new villagers." The village's daily operations depend on external personnel, rendering the village's vitality somewhat performative. Should projects conclude or funding cease, the village risks falling into dormancy.

Physically, many traditional village houses are in disrepair or have even collapsed due to long-term vacancy. Restoration projects, such as the "Lai Chi Wo Hakka Life Experience Village," progress slowly because of fragmented land ownership, high costs, and regulatory restrictions. The deterioration of cultural assets and physical infrastructure threatens the village's integrity as a "living cultural heritage." Moreover, there is a lack of systematic documentation and preservation for intangible cultural heritage, increasing the risk of cultural loss. For instance, traditional Hakka songs and dances are rarely seen or heard as the number of knowledgeable bearers decreases.^[3]

3.3 Institutional and Policy Constraints on Rural Development

Hong Kong's highly urbanized regulatory framework poses distinct challenges to rural growth. On the one hand, prevailing rules about construction, fire safety, sanitation, and business licensing are mainly tailored for high-density city environments, rendering them ill-suited for rural areas. For example, small village houses can't satisfy urban fire safety standards, and traditional building frameworks can't bear heavy firewater tanks, leading to sluggish and expensive legalization of

homestays. On the other hand, scattered private land ownership further complicates comprehensive planning and development coordination, restricting the nurturing of large-scale and sustainable rural industries. Additionally, the bureaucratic steps involved in securing permits and approvals for rural development initiatives are often lengthy and complicated, further impeding advancement. For instance, the procedure of requesting a change in land use from agricultural to commercial ends can last several years, involving numerous government bodies and voluminous paperwork^[4].

3.4 Financial Instability and Weak Internal Revenue Mechanisms

In interviews, a local resident said projects in Lai Chi Wo are short-lived, usually ending in a few years. With an unfavorable environment and a declining population, the village's future is uncertain. The success of the village revitalization plan depends on a diversified funding network. Lai Chi Wo has explored farmland restoration, product processing, farmer's markets, ecotourism, homestay rentals, and established organizations like "HakkaHome" for Hakka life experience programs. However, these activities are small-scale, semi-public, or subsidy-dependent, lacking a self-sustaining commercial cycle^[2].

Currently, most revitalization funding is in the form of "project grants" with 1-3 year cycles, tied to specific objectives. When projects end or funders' strategies change, the funding chain may break, pressuring daily maintenance, staff salaries, and activity organization. According to the Hong Kong Countryside Conservation Office, Lai Chi Wo's funding projects were concentrated from 2021 to 2023, and post-2025 support needs more social funding (Figure 3.1). The stability of the revitalization funding depends on external conditions and funders' willingness, lacking institutional guarantees. This heavy external-funding reliance is a key financial bottleneck for sustainable development. Moreover, the lack of a clear rural development financial policy framework in Hong Kong makes long-term planning and infrastructure and human capital investment difficult for villages like Lai Chi Wo^[5].

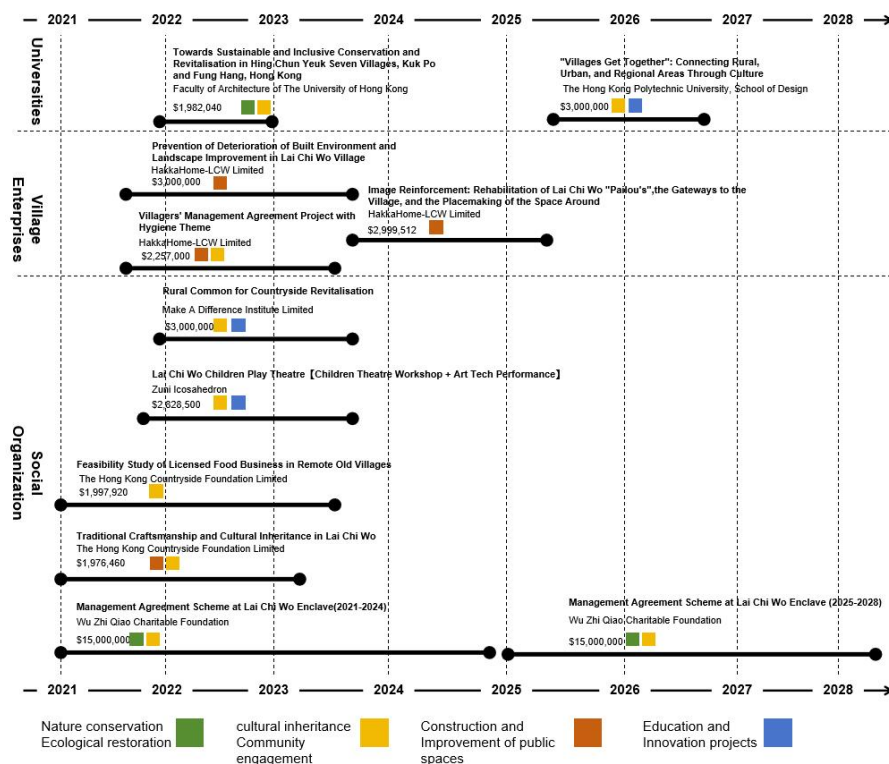


Fig 3.1 A Review of Funded Projects in Lai Chi Wo Village from 2021 to 2028
(Compiled based on the official website of the Hong Kong Countryside Conservation Office)

3.5 Strategic Approaches

The population decline in Lai Chi Wo is inevitable due to its transition from production to consumption and leisure. Efforts should focus on attracting new groups, such as tourists and

creative entrepreneurs, to drive development. Limited accessibility, once a drawback, now protects the village's rural character and enables sustainable tourism. Basic services can be transformed into "cultural experience" products using the "the sixth industrialization of agriculture" model. Culture is central to rural revitalization, and combining cultural transmission with industrial development can turn cultural elements into experiential products, preserving heritage.

Institutional bottlenecks and fragmented land ownership hinder collective action. A collaborative governance platform involving the government, society, market, and villagers can address these issues. The government should provide policy support, while social organizations coordinate private property and guide external capital. This framework leverages Lai Chi Wo's strengths, supporting post-productivism and the agriculture-tourism integration model. Its cross-border features and Hakka culture can attract visitors from Shenzhen and Hong Kong, developing products for the Greater Bay Area's cultural and tourism market.

4. Development Recommendations and Prospects: A Systematic Approach Based on the Three-Tiered Theoretical Framework

4.1 Policy and Institutional Innovation from a Post-productivist Perspective

The Hong Kong government has the potential to shift from a stance of passive non-intervention to that of an active enabler by engaging in top-level design, innovating land use, constructing financial mechanisms, and optimizing governance. This transformation would create a more favorable institutional environment for rural values.

Cross-border cooperation between Shenzhen and Hong Kong is crucial for the joint development of tourism in the Sha Tau Kok area. Implementing a "first-line liberalization and second-line control" management approach can enhance checkpoint efficiency and integrate electronic information systems, allowing for free shopping on Shenzhen-Hong Kong shopping streets while maintaining stringent border controls. Consolidating the current fragmented rural development policies into a cohesive strategy can break down departmental barriers and offer tailored guidance for individual villages.

Innovations in the land system, such as adopting the conservation easement model and establishing a "Rural Revitalization Experimental Zone", can streamline land-use approval processes and safeguard ecological and agricultural functions. By optimizing the governance model through the establishment of a scientific performance evaluation system that includes both economic and ecological indicators, the effective use of funds and resources can be ensured, providing valuable references for future investors.

4.2 Industrial Pathway of Agri-Cultural-Tourism Integration

The core of Lai Chi Wo's revitalization is not traditional agricultural production but the activation of the village's ecological and cultural functions. Project initiators focus not on crop yield but on cultural heritage, ecological protection, and community building. Villagers' initial "return to roots" sentiment and continuous maintenance of houses are driven by strong cultural identity and emotional attachment rather than economic motives. This demonstrates that in postmodern society, the value of rural areas has transcended material production, transforming into new functions that provide psychological comfort, cultural belonging, and ecological leisure (Table 4.1).

4.2.1 Ecological Upgrade of the Primary Industry: The Foundational "1"

The revitalization of Lai Chi Wo started with agriculture, focusing on the ecological and quality-oriented upgrading of the primary industry rather than just restoring production. Under the sustainable development concept, the project set ecological agriculture principles from the start, abandoning chemical fertilizers and pesticides and using natural farming methods. This restored soil vitality, protected biodiversity, produced safe agricultural products, and restored rural ecological functions, laying a foundation for ecotourism. Moreover, the land cultivation project revived local traditional crops like coffee, turmeric, and shiitake mushrooms beyond rice cultivation, providing unique raw materials for subsequent processing and experiential activities.

4.2.2 Experiential Extension of the Secondary Industry: Small-Scale and Refined "2"

In Lai Chi Wo, the secondary industry has adopted a small-scale, hands-on model that integrates processing with cultural and tourism experiences, rather than pursuing large-scale industrial processing. Visitors can participate in activities such as making Hakka tea cakes, grinding coffee, and pickling agricultural products, transforming the "processing" stage into a high-value service product with cultural and educational value. Given the village's geographic limitations, future development can leverage online channels to run e-commerce stores and establish digital platforms for rural revitalization projects. These platforms can serve as direct marketing channels between producers and consumers and facilitate skill-matching between urban professionals and rural needs, thereby improving resource allocation efficiency.

4.2.3 Deep Integration of the Tertiary Industry: Value Multiplication "3"

The tertiary industry is the core for value multiplication in Lai Chi Wo, integrating agriculture, ecology, culture, and education into diversified services. In cultural tourism, the "Hakka Life Experience Village" is a core product, offering a one-stop immersive experience. Visitors become active participants in rural life. In ecological education, Lai Chi Wo, with its good environment and biodiversity, is an important nature-education site. Professional ecological tours spread conservation knowledge. In community building, the project organizes markets, festivals, and volunteer activities to create an urban-rural interaction platform. Urban residents' participation benefits rural development and gives them a sense of satisfaction and belonging. (Table 4.1).

Table 4.1 The Practice of Agricultural Sixth Industrialization in Lai Chi Wo

Domain	Lai Chi Wo practice
1. Produce	Restore wasteland to grow crops in an eco-friendly way and fight off invasive species.
2. Process	New villagers make handmade ice cream from flowers they grow; process turmeric into branded turmeric powder; and plan to turn the coffee plantation into Hong Kong's largest coffee growing base.
3. Serve	Ferry routes are opened; weekend markets are held to sell agricultural products and cultural and creative goods; homestays provide accommodation and experience services.
4. Wellness	The homestay project provides places for tourists to relax; old and new villagers settle down in the village and enjoy the slow pace of rural life.
5. Education	Organize farmer training courses; provide guided tours for tourists; and carryout community-building activities to engage urban residents.
6. Inherit	Restore ancient village houses, ancestral halls and temples; maintain a traditional blessing ceremony once every ten years; position the village as a "living Hakka architecture museum".

4.3 Long-Term Operation through Collaborative Governance

Lai Chi Wo has the potential to draw back villagers, their descendants, and new residents to reestablish its inherent order through a multi-stakeholder collaboration involving the government, social organizations, villagers, and non-profits. This method stimulates rural vitality, encourages community involvement, and preserves cultural heritage. Conflicts between old and new villagers can be managed through institutional design and negotiation mechanisms. Fundamental infrastructure enhancements, such as transportation and healthcare, are crucial for attracting talent and ensuring sustainable rural development. Government investment can be secured through the Hong Kong Countryside Foundation, which fosters ecological conservation and rural revitalization by facilitating collaboration among non-profits, government agencies, and villagers to support projects. The village should also consider the "1×2×3" Farm–Culture–Tourism model to decrease reliance on external investment, generate economic benefits, diversify funding sources, and ensure long-term project sustainability.

Revitalization projects in Lai Chi Wo should advance incrementally and integrate seamlessly to prevent "use-and-abandon" practices. Projects need to be interconnected, forming a systemic

governance pathway that enhances sustainability, prevents resource waste, and alleviates fragmented management. Lai Chi Wo should utilize village enterprises and social organizations, such as the Hong Kong Suburban Fund, to connect villagers with government stakeholders. These organizations provide professional support, coordination, and communication, promoting scientific and transparent project execution. Standardizing village enterprises improves governance efficiency and ensures the smooth operation of revitalization initiatives. The integration of projects and the standardization of village enterprises will further support the village's long-term success and serve as a model for other rural areas facing similar challenges.

5. Conclusion

This research investigates the incorporation of ecological, agricultural, and cultural aspects within border villages of the Guangdong–Hong Kong–Macau Greater Bay Area, with Lai Chi Wo serving as a case study. Utilizing a post-productivist framework and the Farm–Culture–Tourism integration model, the study explores the transformation of values and sustainable development. It reveals that post-productivism redefines resources as key assets, enabling functional shifts. The amalgamation of agriculture, culture, and tourism, coupled with the "sixth industrialization of agriculture" approach, successfully extends the value chain and establishes an ecosystem. Collaborative governance, which includes the government, society, market, and villagers, tackles collective action problems and sets a benchmark for governance innovation. Nevertheless, challenges persist, including restricted access and inadequate infrastructure. Future progress should concentrate on innovation, improved industrial paradigms, and refined governance to foster endogenous development and support rural revitalization in the Greater Bay Area.

References

- [1]龚卫航,何深静,朱金.多主体参与乡村多元价值转型:以香港庆春约为例[J].公共艺术,2022,(03):14-19.
Gong Weihang, He Shenjing, Zhu Jin. Multi-Participants and Plural Values for Rural Transformation: A Case Study of the Hing Chun Yeuk, Hong Kong[J].Public Art,2022,(03):14-19.
- [2]关鑫;姜文来;信军;段玉权;贾伟;李芸;钟葵;候艳林;栗欣如;冯欣.乡村振兴背景下农业“新六产”理论与形成路径研究.中国农业资源与区划 2020, 41 (9), 19–27.
Guan, X., Jiang, W., Xin, J., Duan, Y., Jia, W., Li, Y., Zhong, K., Hou, Y., Li, X., & Feng, X. Research on the Theory of Agricultural "New Sixth Industry" and Its Formation Path under the Background of Rural Revitalization. Chinese Journal of Agricultural Resources and Regional Planning, 2020, 41 (9), 19–27.
- [3]胡一诺.多元主体共建的大都市远郊乡村振兴路径探索——以香港荔枝窝村为例.国际城市规划 2024. 1-17.
Hu YINUO. Rural Revitalization in Exurban Zone of metropolitan: A Case Study of Lai Chi Wo, Hong Kong. Urban Planning International, 2024, 1-17.
- [4]姜乃心;黄一如.多元主体共治的香港乡村复兴研究——以大澳和荔枝窝为例.建筑与文化 2025, No. 1, 156–159.
Jiang Naixin, Huang Yiru. The Rural Revitalization of Hong Kong from the Perspective of Collaborative Governance: Taking Tai O and Lai Chi Wo as Examples. Culture & Heritage, 2025, No. 1, 156–159.
- [5]雷明;王钰晴.交融与共生:乡村农文旅产业融合的运营机制与模式——基于三个典型村庄的田野调查.中国农业大学学报(社会科学版) 2022, 39 (6), 20–36.
Lei, M., & Wang, Y. Blending and Symbiosis: The Operational Mechanism and Model of Agri-cultural-tourism Integration in Rural Areas—Fieldwork Based on Three Typical Villages. Journal of China Agricultural University (Social Sciences), 2022.39(6), 20–36.
- [6]李晨曦;何深静.后生产主义视角下的香港乡村复兴研究.南方建筑 2019, No. 6, 28–33.
LI Chen-xi, HE Shen-jing. Rural Revitalization in Hong Kong: A Post-productivism Perspective. South Architecture, 2019, No. 6, 28–33.
- [7]沈伟民.社会组织参与乡村振兴管理的法律依据与规制研究.中国管理信息化 2019, 22 (24), 169–170.
Shen, W. Research on the Legal Basis and Regulation of Social Organizations' Participation in Rural Revitalization Management. China Management Informationization, 2019. 22(24), 169–170.
- [8]许欢科;滕俊磊.乡村振兴中新乡贤培育的障碍及其对策.南京邮电大学学报(社会科学版) 2019, 21 (1), 35–42, 67.
Xu, H., & Teng, J. Obstacles and Countermeasures for Cultivating New Local Elites in Rural Revitalization. Journal of Nanjing University of Posts and Telecommunications (Social Science), 2019.21(1), 35–42, 67.
- [9]周霖;田喆;朱彩飞.等.乡村振兴背景下农村生态基础设施实施路径[J].资源节约与环保,2023,(06):141-145.
Zhou, L., Tian, Z., Zhu, C., et al. Implementation Path of Rural Ecological Infrastructure under the Background of Rural Revitalization. Resource Economization & Environmental Protection, 2023(06), 141-145.
- [10]香港大学社会科学学院策动永续发展坊.荔枝“永续荔枝窝”四年回顾及展望[M].香港:香港大学社会科学学院策动永续发展坊,2017.
HKU Faculty of Social Sciences Centre for Civil Society and Governance. Lai Chi Wo: Four-year Review and Outlook of the Sustainable Lai Chi Wo Project.[M].Hong Kong: HKU Faculty of Social Sciences Centre for Civil Society and Governance.2017.